Regulation of the motion gain

2026-10-21

1 Regulation law (normal direction $\bar{w}$; 1019 Part III, flat wall $\bar{w} = z$)

Regulation law

$$\Delta\bar{w}_d[k] = \frac{T_s}{\tau_a}\,\frac{\bar{a}_d - \hat{\bar{a}}_N[k]}{\hat{\bar{a}}'_N[k]}$$

$$\bar{w}_d[k+1] = \bar{w}_d[k] + \Delta\bar{w}_d[k]$$

Motion gain under the regulation law ($\hat{\bar{a}}_N = \bar{a}_N$, $\hat{\bar{a}}'_N = \bar{a}'_N$, first order in $\Delta\bar{w}_d$)

$$\bar{a}_N[k+1] = \bar{a}_N[k] + \bar{a}'_N[k]\,\Delta\bar{w}_d[k]$$

$$\bar{a}_d - \bar{a}_N[k+1] = \lambda_a\left(\bar{a}_d - \bar{a}_N[k]\right)$$

$$\lambda_a = 1 - T_s/\tau_a$$

2 Three axes: regulation along $\hat{\mathbf{w}}$, control of $\bar{u}_d$, $\bar{v}_d$

Desired position along the wall in the estimated local coordinate, relative to the start

$$\bar{u}_d[k] = \hat{\mathbf{u}}[k]^{\mathsf{T}}\left(\bar{\mathbf{p}}_d[k] - \bar{\mathbf{p}}_d[0]\right)$$

$$\bar{v}_d[k] = \hat{\mathbf{v}}[k]^{\mathsf{T}}\left(\bar{\mathbf{p}}_d[k] - \bar{\mathbf{p}}_d[0]\right)$$

Control law

$$\Delta\bar{w}_d[k] = \frac{T_s}{\tau_a}\,\frac{\bar{a}_d - \hat{\bar{a}}_N[k]}{\hat{\bar{a}}'_N[k]}$$

$$\Delta\bar{u}_d[k] = -\frac{T_s}{\tau_u}\,\bar{u}_d[k]$$

$$\Delta\bar{v}_d[k] = -\frac{T_s}{\tau_u}\,\bar{v}_d[k]$$

$$\bar{\mathbf{p}}_d[k+1] = \bar{\mathbf{p}}_d[k] + \hat{\mathbf{V}}[k]\begin{bmatrix}\Delta\bar{u}_d[k]\\ \Delta\bar{v}_d[k]\\ \Delta\bar{w}_d[k]\end{bmatrix}$$

Under the control law ($\hat{\mathbf{V}}[k+1] = \hat{\mathbf{V}}[k]$)

$$\bar{a}_d - \bar{a}_N[k+1] = \lambda_a\left(\bar{a}_d - \bar{a}_N[k]\right)$$

$$\bar{u}_d[k+1] = \lambda_u\,\bar{u}_d[k]$$

$$\bar{v}_d[k+1] = \lambda_u\,\bar{v}_d[k]$$

$$\lambda_a = 1 - T_s/\tau_a$$

$$\lambda_u = 1 - T_s/\tau_u$$

Change of $\hat{\mathbf{V}}$ between two steps (first order in $\Delta\hat{\theta}_y$, $\hat{\theta}_x = 0$)

$$\bar{u}_d[k+1] = \lambda_u\, \bar{u}_d[k] - \Delta\hat{\theta}_y[k]\; \hat{\mathbf{w}}[k]^{\mathsf{T}}\big(\bar{\mathbf{p}}_d[k+1] - \bar{\mathbf{p}}_d[0]\big)$$

Setting

$$\tau_a = 2\,\mathrm{s}, \qquad \sqrt{p_\theta[0]} = 10^\circ, \qquad q_\theta = 0$$

3 Hold, approach, scan

Angle information per sample (angle rows of the output, random component)

$$I_\theta[k] \approx \frac{\left(\bar{a}_T[k] - \bar{a}_N[k]\right)^2}{2.5\,\bar{a}_T[k]\,\bar{a}_N[k]}$$

Process noise of the angles

$$\begin{aligned} q_\theta[k] &= 0, && t < t_{\text{scan}} \\ q_\theta[k] &= p_{\text{fix}}^2\, I[k], && t \geq t_{\text{scan}} \\ I[k] &= 1/p_\theta[k] - 1/p_\theta^f[k] \end{aligned}$$

Desired position

$$\begin{aligned} \Delta\bar{w}_d[k] &= 0, && t < t_{\text{hold}} \\ \Delta\bar{w}_d[k] &= \frac{T_s}{\tau_a}\,\frac{\bar{a}_d - \hat{\bar{a}}_N[k]}{\hat{\bar{a}}'_N[k]}, && t \geq t_{\text{hold}} \\ \Delta\bar{u}_d[k] &= \Delta\bar{u}_{\text{scan}}, && t \geq t_{\text{scan}} \end{aligned}$$

$$\bar{\mathbf{p}}_d[k+1] = \bar{\mathbf{p}}_d[k] + \hat{\mathbf{V}}[k] \begin{bmatrix} \Delta\bar{u}_d[k] \\ 0 \\ \Delta\bar{w}_d[k] \end{bmatrix}$$

Setting

$h[0] = 15\,\mu\text{m}$, $t_{\text{hold}} = 4\,\text{s}$, $t_{\text{scan}} = 11\,\text{s}$, $\sqrt{p_{\text{fix}}} = 2^\circ$, $\Delta\bar{u}_{\text{scan}} = 0.011$, no control of $\bar{u}_d, \bar{v}_d$

Fig. 3.1 Angle while holding at 15 μm

20° plane wall, held 15 s at a 15 μm gap; columns: no dither, dither without ramp, dither with 2 s ramp, dither with 5 s ramp (circle of radius 3 μm at 8 Hz in the estimated tangent plane); rows: angle errors, $\sqrt{p_{\theta_y}}$; seeds 1–3

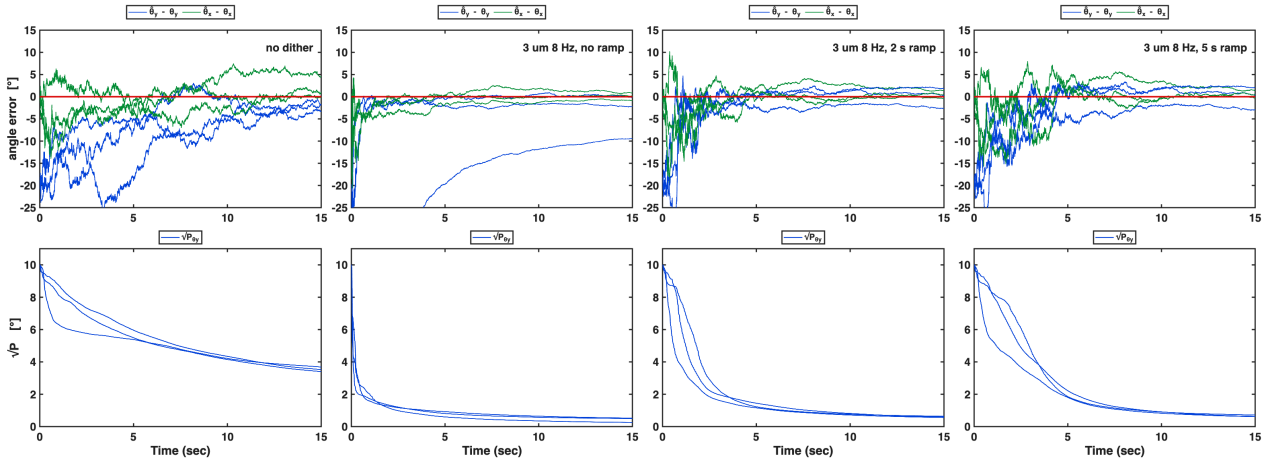


Fig. 3.2 Hold, approach, scan on the 20° plane wall

hold 4 s with the dither (2 s ramp), regulation from 4 s, scan from 11 s; side view in the measurement coordinate with the true normal through the start (light red dashed); seed 1

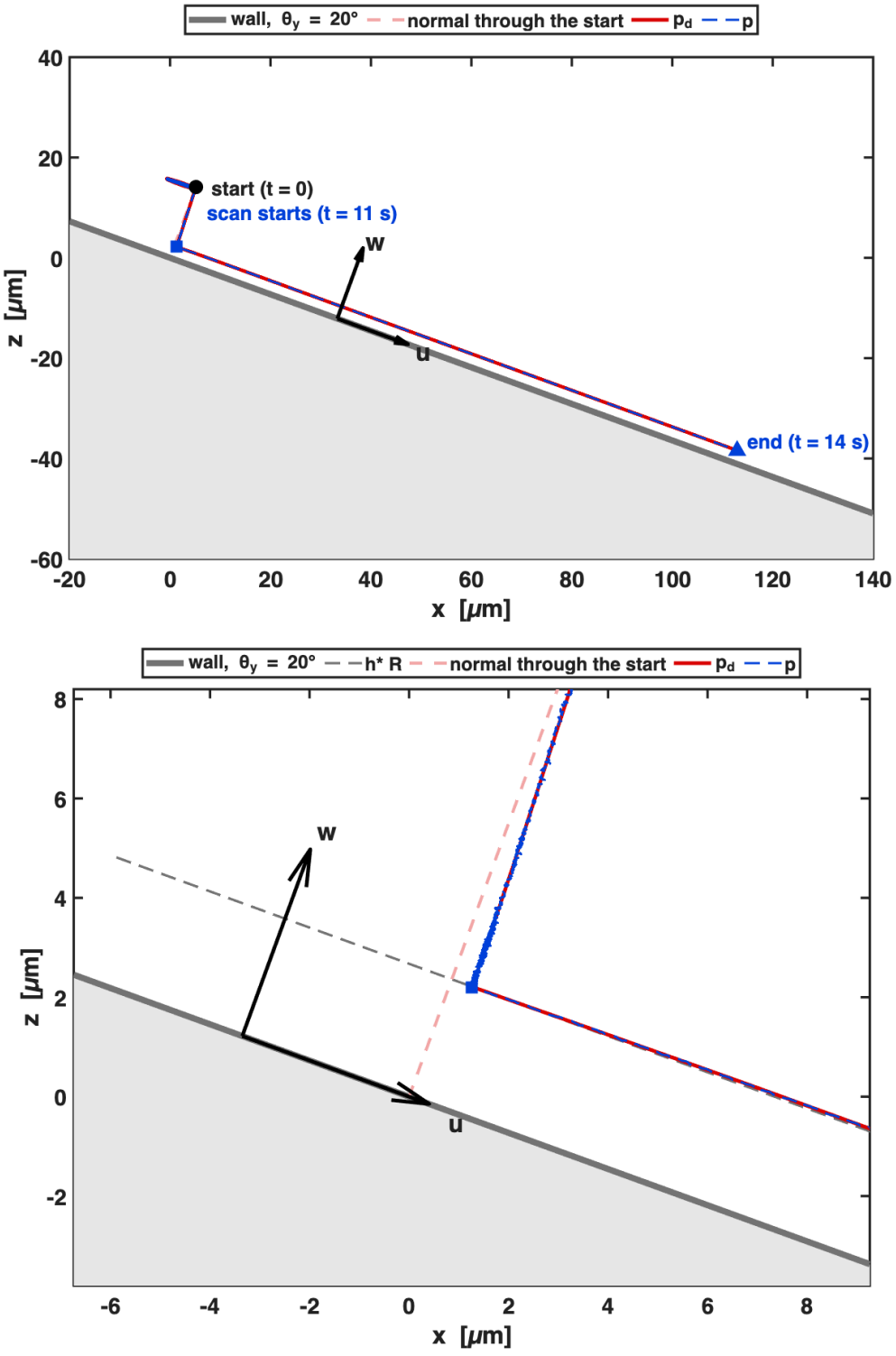


Fig. 3.3 Estimates of Fig. 3.2

rows: $\bar{a}_N$ with $\bar{a}_d$, $\bar{a}_T$, θ_y and θ_x with $\pm\sqrt{p_\theta}$; seed 1

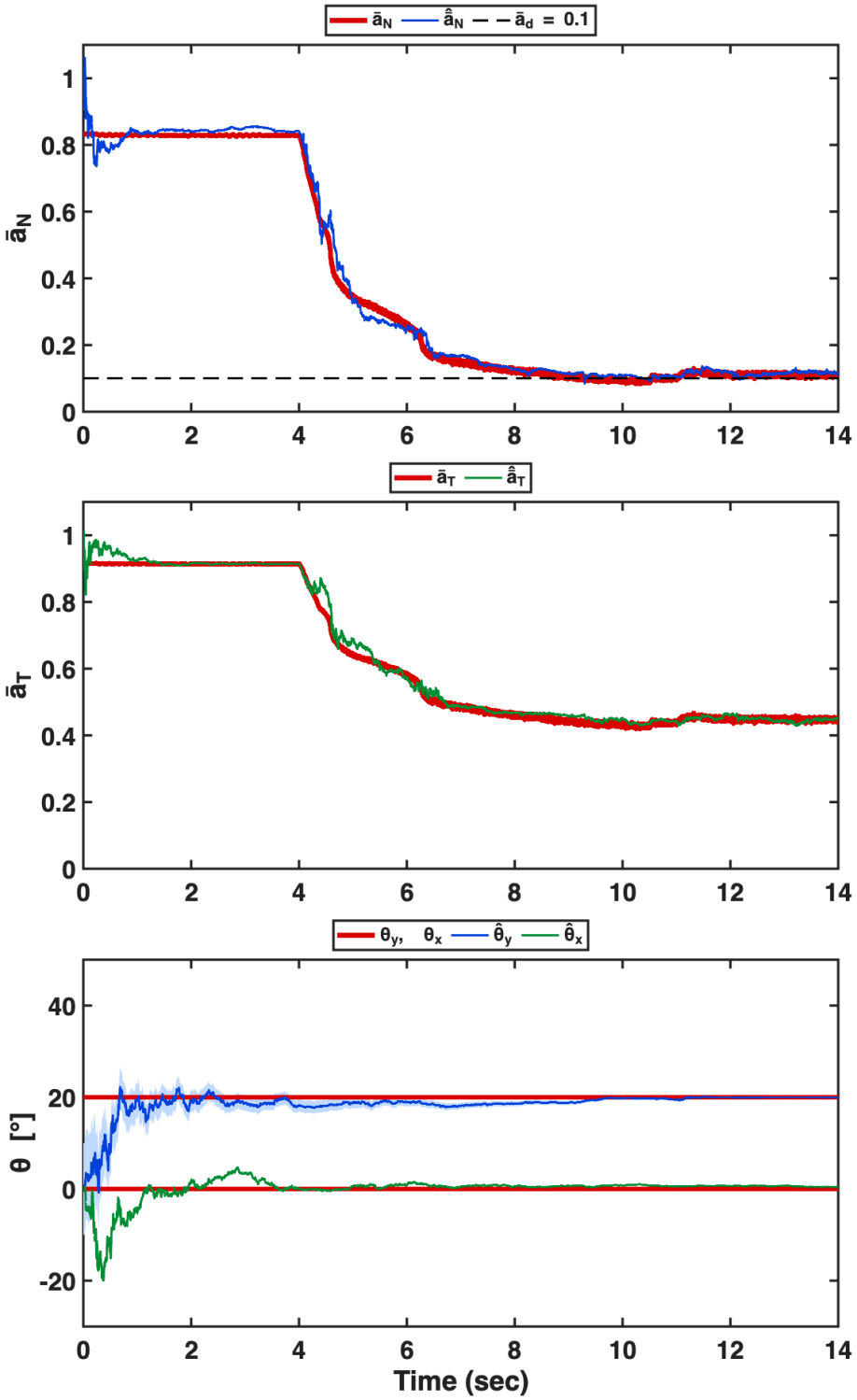


Fig. 3.4 Hold, approach, scan on the bend wall

flat, 20 μm bend, 20° plane (downslope azimuth 45°); left: 10-20 run with the gate and prior 30°; middle: start 22.2 R, p_{fix} from the scan; right: Section 3 setting; rows: θ, h, $\bar{a}_N$; seed 1

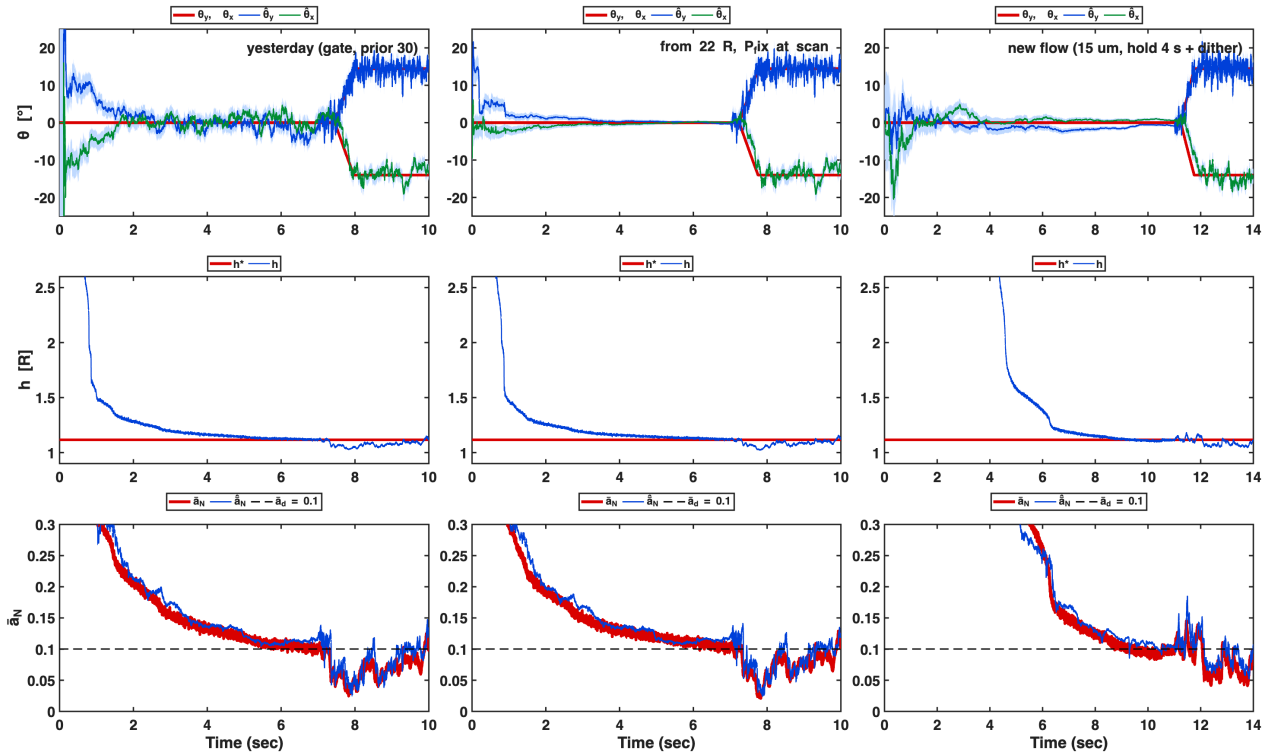


Fig. 3.5 Side view of the right column of Fig. 3.4

wall drawn at the scan line; lower panel: the bend with the gap h^*R; seed 1

